

[This manuscript has been accepted for publication in *Nature*.]

Title: Large language models can predict the results of social science experiments

Authors: Ashwini Ashokkumar^{1*†}, Luke Hewitt^{2,3*†}, Isaias Ghezae¹, Robb Willer^{2*}

Affiliations:

¹Department of Psychology, Harvard University; Cambridge, 02138 USA.

²Department of Sociology, Stanford University; Stanford, 94305 USA.

³Transluce; San Francisco, 94111 USA.

*Corresponding authors. Emails: ashwiniashokkumar@g.harvard.edu, luke@transluce.org, willer@stanford.edu

†These authors contributed equally. Their names are listed in a random order.

Abstract: There is growing interest in how large language models (LLMs) can advance social and behavioral science [1–5]. Prior work has assessed LLMs’ ability to predict survey responses [6–9], but less is known about whether they can predict the outcomes of social science experiments [10], particularly those absent from training data. Here, we built an archive of 70 pre-registered, nationally representative, U.S. survey experiments, involving 469 experimental effects and 119,330 participants. We prompted an LLM to simulate how representative samples of Americans would respond to experimental stimuli, then inferred treatment effects by comparing simulated responses across conditions. Predictions derived from GPT-4, whose training-data cutoff predated the publication of many studies in our archive, were strongly correlated with actual treatment effects, achieving accuracy similar to pooled human forecasts. Correlations remained high for studies not published or publicly posted by the model’s training-data cutoff date, and for predictions from prominent open-weight models. Despite high correlations, predictions systematically overestimated effect sizes. In a secondary archive of 15 megastudies featuring 606 effects, correlations were lower but comparable to pooled expert forecasters. To assess implications for scientific practice, we surveyed 460 social scientists about likely uses and perceived risks, and used our archives to assess several applications (pilot testing, intervention selection, identifying effects needing replication) and risks (bias, misuse). Together, these results suggest LLMs can augment experimental methods in science and practice while raising important considerations for responsible use.

Main text:

Introduction

Large language models (LLMs) - machine-learning models trained on vast language corpora - have been shown to mimic human behaviors and abilities, such as social interaction, moral judgment, and persuasion [11, 12, 0, 13–16]. With LLMs increasingly able to simulate human language use, pressing questions have emerged about whether and how these models can be used to support social and behavioral science [1, 2, 17]. Here, we explore whether LLMs can be leveraged to accurately predict the results of behavioral experiments, a capacity that could have important benefits for building scientific theories and behavioral interventions. We examine whether leading publicly available LLMs can be used to predict experimental effects observed in (a) a large archive of well-powered, nationally representative experiments conducted through the NSF-funded Time-Sharing Experiments in the Social Sciences (TESS) program [18], or included in a recent meta-analysis of replication studies [19], and (b) an archive of 15 recent large-scale, many-treatment experiments (or “megastudies”) assessing the effects of multiple treatments on socially desirable outcomes, together representing a diversity of fields (e.g., social psychology, political science, sociology, public policy). We prompt the LLM to simulate the responses to experimental stimuli of demographically diverse samples of Americans and then compare average responses across different experimental conditions to generate LLM-predicted experimental effect sizes, which we then compare to observed effects. We assess the accuracy of LLM-derived predictions, systematically benchmarking the capability to predict treatment effects observed in survey experiments. We also assess several potential scientific applications of the ability to predict experimental results using AI, as well as limitations and risks, to assess the current value and limitations of LLM-derived experimental predictions.

The ability to predict social science experimental results could have valuable implications for basic and applied social science [3, 20]. AI-based pilot studies could augment human samples [21], helping researchers refine research ideas, identify promising hypotheses, and better estimate effect sizes to determine needed sample sizes. AI predictions could also help identify published studies in need of replication. Applied researchers and policymakers could use LLMs to assess the effectiveness of messages encouraging desirable behaviors (e.g., public health behaviors). For these purposes, scientists at present rely on their own intuitive judgments of expected results, or collect expert and lay forecasts [22–25], which is often time-consuming and costly; LLM-based methods may offer a low-cost alternative that improves these predictions. Across these applications, LLM-derived predictions are best viewed as supplementary tools that assist— not replace—the study of human responses [4, 5, 26–28].

Here, we first investigate the capacity for LLMs to be used to accurately forecast experimental results by simulating human responses in representative survey experiments – studies where an experimental treatment is administered and outcomes are measured within surveys conducted on probability samples of a larger population. Recent research has focused on prompting or

finetuning LLMs to simulate human responses to survey questions with variable success [6, 8, 9, 14, 29, 30]. Instead, we focus on evaluating LLMs’ ability to predict *experimental effects* - differences in outcome levels caused by randomized treatments, which are central to causal inference across the social sciences. Unlike simulating cross-sectional surveys, predicting experimental results requires accurately capturing how responses to outcomes vary across subtly different conditions [31]. Some prior work has used LLMs to predict experimental results [10, 20, 32]; for instance, one study found that LLMs can forecast the results of well-known economics experiments [10]. However, rigorously assessing LLMs’ ability to predict *novel* treatment effects is a challenge, as it requires a large set of diverse and well-powered experiments that are unlikely to appear in the training data. We assess LLM performance across two archives featuring experiments from multiple fields and conducted in varied settings.

While promising, the use of LLMs to predict experimental results likely has limitations, requiring rigorous assessment of the method’s scope and attention to possible risks. First, given concerns that LLMs’ responses may be biased against groups with less access to the internet, or which are historically underrepresented and/or misrepresented in news or other media [4, 5, 7, 8], we assess whether the accuracy of LLM predictions varies across demographic groups in the U.S. Further, because widespread availability of technology that accurately predicts experimental results might carry societal risks, we test whether LLMs can be employed to design harmful interventions, such as identifying content that can mislead the public. Finally, we present findings from a survey of a large and diverse sample of social scientists assessing their interest in using validated AI-driven predictive tools in their research, as well as their perceptions of associated risks.

Study Overview

Here, we examine whether LLMs can be leveraged to accurately predict the magnitude of social science experimental effects conducted in the U.S. We built a large, multidisciplinary testing archive consisting of 50 survey experiments conducted via the NSF-funded TESS project [18] from 2016 to 2022 on nationally representative probability samples. We supplemented this with 20 experiments from a recent replication project [19]. In addition to being high quality (high-power, peer-reviewed, preregistered, with open materials/data) and diverse (77 researchers across fields, different types of treatments and outcomes), a large number of experiments were not published, nor posted publicly, by the end of GPT-4’s training data window, allowing us to specifically test for LLMs’ predictive capacity on *novel* experiments. To address some of this archive’s limitations, we also analyze a secondary test archive below.

Our approach is illustrated in Extended Data Figure 1. We prompted the LLM with (a) a randomly drawn introductory message (e.g., “You will be asked to predict how people respond to various messages”) including a brief description of study setting, (b) a hypothetical research participant’s demographic profile (gender, age, race, education, ideology, and partisanship) drawn from a large nationally representative sample based on population benchmarks, (c) the

experimental stimulus, and (d) the outcome question with response scale and labels. We then asked the LLM to estimate the described participant’s response [6, 10, 28]. For each condition and outcome, we generated 120 prompts; for each prompt, we sampled five model responses and averaged them to produce one prompt-level simulated response, then used an ensemble approach averaging across prompts to reduce idiosyncratic responding to any single prompt format. [33–36]. While we focus on GPT-4, we also report the main results from three open-weight models—Gemma-3, GPT-OSS, and DeepSeek-V3—to address reproducibility concerns associated with proprietary models.

We used a consistent analytical approach across experiments that does not rely on study-specific models. For all experimental contrasts, we computed predicted treatment effects based on LLM responses, and then following prior work [19, 25, 37–41], we correlated LLM-predicted effects with corresponding effects from the original studies (r). Because raw correlations do not account for sampling variability associated with original effect size estimates (i.e., correlations are biased downward when original effects are estimated imprecisely) [41], we also report adjusted correlations (r_{adj} ; See *Method*), which generally has a small impact when the original effects are precisely estimated, but can have a larger impact when effect sizes are less precisely estimated, such as in small samples. Further, correlations reflect how well predictions discriminate among effects of varying magnitudes but do not indicate whether predictions over- or underestimate effect size. To assess how calibrated predictions are, we also report coefficients from OLS models regressing original effect sizes on LLM-derived predictions, as well as the adjusted Root Mean Squared Error (i.e., $RMSE_{adj}$; See *Method*) which quantifies the average magnitude of prediction errors. Nevertheless, we prioritize correlations because discrimination between effect sizes is valuable for several applications (e.g., selecting effective interventions; [41]), and predictions can be linearly re-calibrated to the correct scale.

Results

Primary test archive

To evaluate whether LLMs can be leveraged to predict experimental effects, we first examined the correlation between LLM-derived effect sizes, and actual estimated effect sizes. In an analysis of 469 effects from 70 experiments, GPT-4-derived predictions were strongly correlated with original effect sizes ($r = 0.85$, 95% CI = [0.81, 0.88]; $r_{adj} = 0.92$, 95% CI = [0.89, 0.94]). As shown in Figure 1B, the correlation improved steadily and markedly across generations of LLMs, between GPT-3 Davinci (with 175B parameters) and GPT-4 (believed to have approximately 1 trillion parameters). Addressing concerns about reproducibility associated with proprietary models, we also find high correlations when using prominent open-weight models. Supporting the validity of our ensemble strategy, the correlation increased when using a larger number of prompts (see Extended Data Figure 2). Additionally, as shown in Figure 1C, LLM-derived predictions achieved high correlations for studies conducted across a range of academic fields.

Despite the high correlation between LLM-derived and true effects, LLM predictions sys-

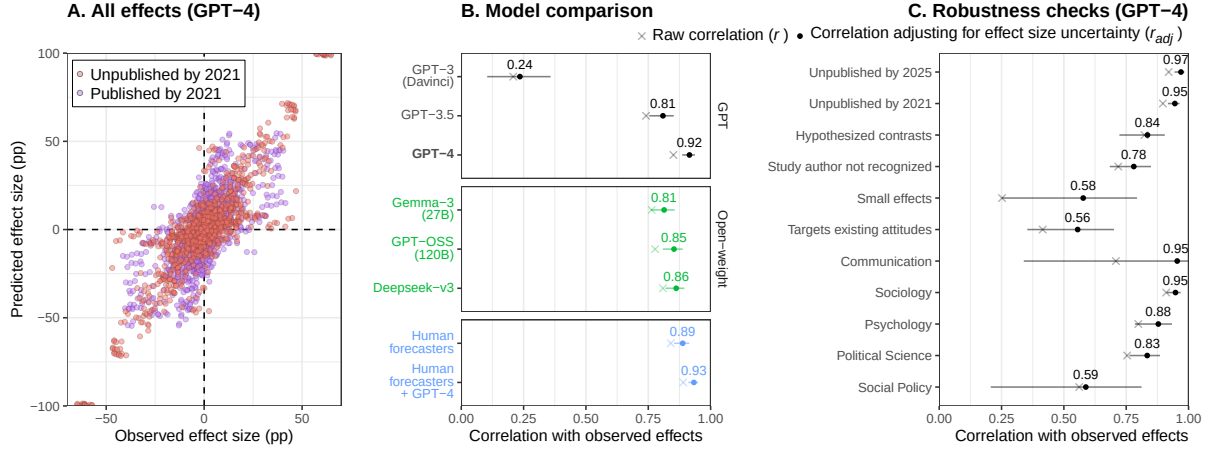


Figure 1: LLM predictions were strongly correlated with original treatment effects in 70 text-based social science survey experiments conducted in the US. (a) GPT-4-derived predictions of effect size were strongly correlated with original treatment effects ($r = 0.85$; $r_{adj} = 0.92$). The figure shows all pairwise contrasts from the primary test archive. (b) The correlation improved across generations of LLMs and remained high when using open-weight models. (c) In robustness check analysis of various subsets of experiments, the correlation of LLM-derived predictions and original effects remained high. Error bars indicate 95% confidence intervals.

tematically overestimated effect sizes ($b = 0.56$ [0.52, 0.60], $p < .001$): LLM-derived effect size predictions were roughly twice as large as the actual effect sizes, yielding a large RMSE of 11.09pp between predicted and true effects ($\text{RMSE}_{adj} = 10.59\text{pp}$; Also see Figure S2, SI). Therefore, when using our approach to predict effect magnitudes, it is important to recalibrate LLM-derived predictions using relevant benchmark data where available as we illustrate below for LLM-based pilot-testing.

To assess the possibility that the model is simply retrieving and reproducing published results from training data, we evaluated predictive accuracy for unpublished studies that were unlikely to appear in the training corpus. We systematically coded whether each study had been published or posted as a preprint and examined predictive accuracy for studies that were not published as peer-reviewed articles or publicly posted preprints before GPT-4's training data cut-off (September 2021; $N = 30$; 252 effects; see *Method*), and thus would not have been part of training data. Inconsistent with this concern, we found a high correlation ($r = 0.90$ [0.86, 0.93]; $r_{adj} = 0.95$ [0.92, 0.96]; $\text{RMSE}_{adj} = 10.22\text{pp}$; see Figure 1C), although the predictions systematically overestimated effect sizes ($b = 0.60$). To account for potential training data leakage beyond the cutoff date, we examined the 19 studies that remained unpublished as of June 2025 (138 effects; $r = 0.92$ [0.88, 0.95]; $r_{adj} = 0.97$ [0.94, 0.98]; $\text{RMSE}_{adj} = 12.06\text{pp}$; $b = 0.61$; see Figure 1C) and studies whose authors the model could *not* guess (see Section 3.1, SI) and found high correlations. Moreover, if LLMs were simply retrieving results from training

data, correlations should be higher for hypothesized effects— those most likely to appear in the literature— but our results ($r = 0.82$ [0.71, 0.90]; $r_{adj} = 0.84$ [0.72, 0.90]; $RMSE_{adj} = 9.38$; $b = 0.53$) do not support this. While we cannot know which effects appear in the proprietary LLM training data, our analysis suggests high correlations for novel effects.

To establish a benchmark for predictive accuracy, we collected effect size forecasts from a large sample ($N = 2,659$) of American laypeople from Prolific and computed the average forecast for each effect (see *Layperson Forecasts*, SI). Averaged forecasts were strongly correlated with original effects ($r = 0.84$ [0.80, 0.87], $r_{adj} = 0.89$ [0.85, 0.91]), and GPT-4 predictions matched this benchmark ($p = 0.118$), where earlier generation models did not (GPT-3.5: $p < .001$; see Figure 1B). Further, similar to LLMs, human forecasters also systematically overestimated effects ($b = 0.58$, [0.53, 0.62], $p < .001$) [42], leading to a large estimated RMSE of 10.57pp ($RMSE_{adj} = 10.04$ pp), similar to GPT-4 ($p = 0.412$). Table S1 in SI depicts forecaster accuracy for the various subsets of experiments shown in Figure 1C.

In further analysis, we examine two explanations for the similar accuracy of human forecasts and LLM predictions: they either provide similar information or independent information. Regression analyses show both LLM-derived predictions ($b = 0.33$ [0.30, 0.36], $p < .001$) and human forecasts ($b = 0.25$ [0.22, 0.28], $p < .001$) are positively and independently associated with true treatment effects, suggesting they offer partly independent information and could be combined to improve accuracy. Indeed, a simple unweighted average of the human and LLM-derived predictions was more strongly correlated with actual effects ($r = 0.89$ [0.86, 0.91]; $r_{adj} = 0.93$ [0.91, 0.95]; $RMSE_{adj} = 8.59$) than human forecasts ($p_r < .001$; $p_{RMSE} < .001$) or LLM predictions ($p_r = 0.002$; $p_{RMSE} = 0.005$) alone, suggesting that - at this time - human-LLM collaboration yields the highest levels of forecasting accuracy.

In summary, we find that LLM predictions are strongly correlated with survey experimental effects conducted in the US, and that this correlation improved across generations of LLMs and is not substantially driven by the LLM retrieving results from its training data. The correlation of LLM predictions matched that of human lay forecasters. Notably, LLM predictions consistently overestimated experimental effect sizes, as did human forecasters. To facilitate the use of this predictive ability for academic research, we built a [web tool](#) that can be used to generate AI-based forecasts of experimental effects.

Secondary test archive

Given growing interest in the causes of normatively desirable attitudes and behaviors, we analyzed a secondary archive of 15 large-scale “megastudies” comparing multiple treatments’ effects on socially valuable outcomes, such as pro-democratic attitudes and flu vaccine uptake. This archive [23, 25, 38–41, 43–51] included 10 survey and five field experiments from multiple disciplines, in total featuring 606 treatment effects estimated from the responses of 4,862,362 original participants (See *Methods* and Table S4 in SI for more details). Notably, several megastudies also collected experts’ (social scientists, and practitioners working in behavioral science organizations) forecasts of likely effect size, providing an additional benchmark for

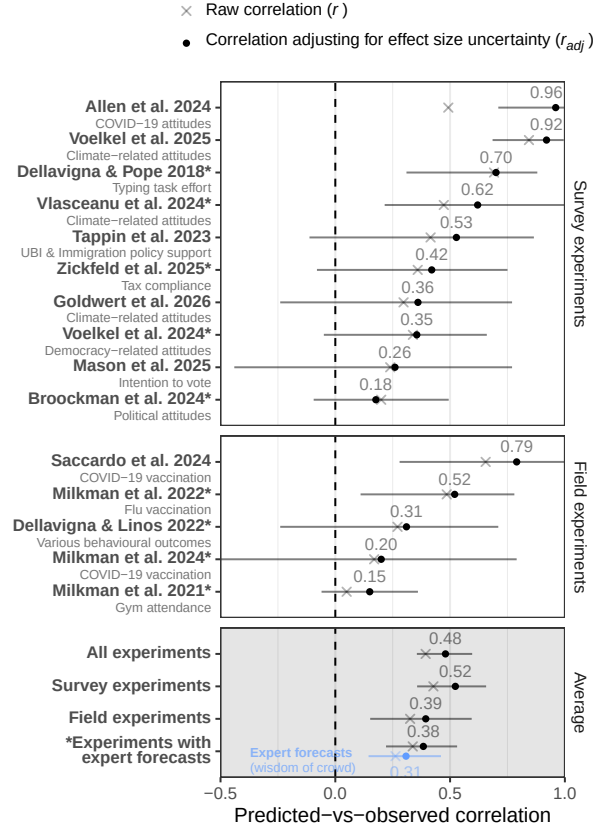


Figure 2: LLM-derived predictions were correlated with treatment effects in megastudies as strongly as were the average expert forecasts. Asterisks indicate studies that included expert forecasts of effect size. Error bars indicate 95% confidence intervals.

evaluating the accuracy of LLM-derived predictions.

This secondary archive of megastudies differs from the primary archive in ways that may affect predictive accuracy. This archive includes experiments with non-text-based treatments (four of 15) and field-based behavioral outcomes (five of 15), which are harder to represent in LLM prompts. Additionally, most survey megastudies in this archive targeted stable attitudes (e.g., climate policy support) and yielded small treatment effects—both harder to predict (see Figure 1c and SI Table S1). Consistent with this reasoning, human forecasts were less accurate in the secondary archive than in the primary archive, despite the former’s forecasters being domain experts.

Because each megastudy compares multiple treatments within the same experimental framework, we first assess correlations of LLM predicted and original treatment effects *within* each megastudy and then meta-analyzed these study-level correlations across studies. Overall, LLM-derived predictions of effect size were positively associated with original effect sizes ($r = 0.39$ [0.27, 0.52]; $r_{adj} = 0.48$, [0.36, 0.60]) but were poorly calibrated in scale and systematically

overestimated effect sizes ($b = 0.34$), as in the primary archive. Across the nine megastudies with expert forecasts, LLM-derived predictions ($r = 0.34$ [0.18, 0.49]; $r_{adj} = 0.38$ [0.22, 0.53]; $b = 0.40$) showed similar predicted–actual correlation to expert forecasts ($r = 0.26$ [0.01, 0.51]; $r_{adj} = 0.31$ [0.14, 0.46]; $b = 0.41$) ($p = 0.30$; Also see Figure S3, SI).

While our archive does not have enough studies to allow statistical comparisons, descriptively we found a stronger correlation for the survey experiment megastudies ($r = 0.43$ [0.28, 0.58]; $r_{adj} = 0.52$, [0.36, 0.66]; $b = 0.35$; see Figure 2) than for the field experiment megastudies ($r = 0.33$ [0.02, 0.63]; $r_{adj} = 0.39$, [0.15, 0.59]; $b = 0.34$). Similarly, the correlation was higher for megastudies testing text-based treatments ($r = 0.45$ [0.31, 0.59]; $r_{adj} = 0.54$, [0.40, 0.66]; $b = 0.41$) than for megastudies testing partially or non-text based treatments (e.g., short videos; $r = 0.24$ [-0.06, 0.56]; $r_{adj} = 0.31$, [-0.01, 0.55]; $b = 0.31$).

Although the correlation is lower in the secondary archive than the primary archive, the same pattern holds for human forecasters. Moreover, isolating comparable effects - survey experiments treating stable attitudes with small effect sizes - yields similar correlations across the archives (see "targets existing attitudes" and "small effects" in Figure 1c). This suggests the apparent gap is compositional, driven by the different features of treatment effects in the two archives. In summary, LLM predictions were substantially correlated with original effects for survey megastudies in our secondary test archive, and exhibited correlations at least as high as those of experts, despite a systematic overestimation of effect sizes.

Scientific uses of LLM predictions

The ability to predict experimental effect sizes using LLMs holds promise for several scientific applications— including conducting AI-based pilot studies, identifying published effects that may need replication, and identifying effective public messaging interventions. Below, we use practically relevant metrics to evaluate the usefulness of LLMs for three scientific applications that a large and diverse sample of social scientists we surveyed reported as likely to use.

LLM-based pilot testing

Researchers conduct pilot studies prior to running large experiments, to anticipate results or improve stimuli and measures. Budget constraints often limit the sample size researchers can collect. We propose that LLMs can reduce the cost of pilot studies, both when used on their own or in conjunction with small human samples. To validate this use, for each experiment in our primary archive, we draw sub-samples of varying sizes— analogous to human pilot samples— and test how well the subsample estimates of effect sizes for originally hypothesized effects predict the corresponding effect sizes estimated from the remaining data (See SI 2.3 and Figure S4). We then compare RMSE of these subsample predictions (i.e., *Human-only pilot studies*) with that of (a) *LLM-only pilot studies* (relying only on LLM-derived predictions), and (b) *LLM+Human pilot studies* (combining LLM-derived predictions with small human pilot samples). To correct for LLM overestimation of effect sizes, we rescale the LLM predictions (see SI 2.3). As

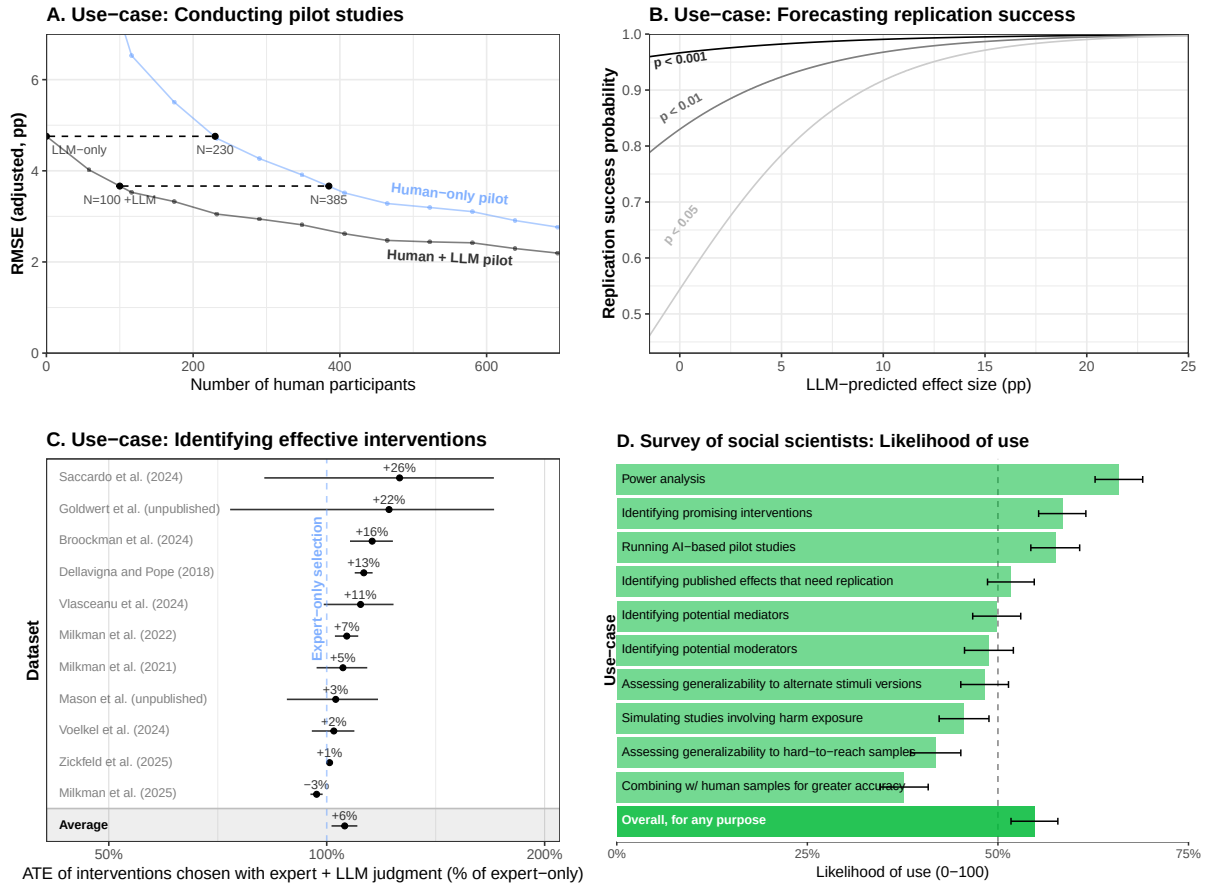


Figure 3: Scientific uses of LLM-derived predictions of experimental results: LLM predictions can be used to effectively (a) supplement human pilot studies at low cost, (b) identify experimental effects that are less robust and need replication, (c) augment expert judgment in identifying top interventions. (d) A survey of 460 social scientists reveals a meaningful likelihood of using LLM-predicted experimental results. In Panel B, the lines depict different significance levels in the initial sample. Error bars indicate 95% confidence intervals.

shown in Figure 3A, LLM-only pilots (costing less than \$1 per study for most models) yield comparable RMSE to human samples of 230 participants (costing approximately \$736; See SI 3.4). Researchers can also combine LLM predictions with estimates from small human pilot samples. For example, combining LLM predictions with estimates from a pilot study of 100 participants (costing approximately \$320 in total), matches the RMSE of samples with 385 participants (costing about \$1232). These findings suggest that LLMs could provide low-cost approaches for pilot tests of causal hypotheses, which may be especially valuable when limited by resource constraints. In practice, researchers can apply this approach by simulating responses under each condition, estimating the treatment effect. Because we observed systematic

overestimation of effect sizes in our archives, researchers should recalibrate predictions using relevant benchmark data when available and otherwise treat absolute effect-size estimates as approximate.

Identifying effects needing replication

If AI predictions can help simulate pilot tests to identify robust effects, they may also help identify effects that are *less* robust and thus less likely to be reproducible in well-powered samples. Current efforts to identify published findings in need of replication are often ad hoc, relying on the judgments of individual researchers. AI predictions could offer a scalable and low-cost approach to complement these efforts. To test this, we take a somewhat similar approach to the assessment of pilot studies above. We first isolate all hypothesized contrasts that were statistically significant in subsamples comprising 20% of the original experimental samples in the primary archive (245.7 participants per contrast, on average), analogous to effects published on the basis of underpowered samples. We then test the same contrasts in the remaining 80% of the samples—analogous to well-powered replication samples—to determine which effects successfully "replicated" and which ones did not. We then ask whether LLM predictions of effect size accurately distinguish between the effects that replicated and the ones that did not. That is, we ask if LLMs can help identify effects that are statistically significant in a small sample but fail to replicate in a second adequately powered sample. In a logistic regression, LLM-derived predictions of effect size significantly predicted replication success ($b = 0.22, p < .01$). A 1pp increase in LLM-derived effect size corresponded to 25.0% increase in the odds of successful replication. LLM predictions remained predictive ($b = 0.27, p < .001$; see Figure 3B) even when controlling for the metrics that are commonly used to identify non-robust effects, such as the original sample size ($b = .0008, p > .05$), effect size ($b = -0.12, p > .05$), and test statistic ($b = 0.02, p < .001$). See Table S2 in SI for additional analyses. Notably, this analysis focuses solely on threats to replicability posed by underpowered samples—and does not account for non-replication due to questionable research practices, or meaningful contextual variation [52, 53]. Another challenge is contamination from training data: published effects may appear in LLM corpora, potentially biasing predictions toward replication. Although our proxy for published effects appearing in LLM training data is imperfect, our findings suggest that LLMs could be useful for open science.

Identifying effective interventions

Another valuable application is identifying effective interventions with positive social impact. Policymakers and researchers often rely on expert intuition and judgment when choosing which of many ideas to test. Using 11 megastudies from our secondary archive – where we have expert forecasts of absolute or relative effect size, we assess whether LLM predictions improve individual experts' ability to select effective interventions. For each megastudy, we identify the top 20% of interventions, that each expert would select if they relied (a) solely on their own

forecast (*expert-only selection*), or (b) the average of their own forecast with LLM predictions (*AI-augmented expert selection*; see SI 2.4). If LLMs add value in intervention selection, the LLM-augmented selection approach would lead to selecting treatments with larger original effect sizes than the expert-only approach. Consistently, treatments selected using the LLM-augmented approach were modestly but significantly more effective, corresponding to 6% larger effects in the original experiments. Moreover, when selecting the *single* most effective intervention, the LLM-augmented approach allows for selection of an intervention that is 8% more effective (See Figure S5, SI). These results suggest that LLMs may provide incremental value for identifying and developing socially beneficial interventions.

Likelihood of use among social scientists

While the above analyses show potential value for LLM-derived predictions across several scientific applications, an important factor in evaluating their scientific utility is the extent to which social scientists would adopt these tools. We surveyed 460 social scientists across career stages (professors, postdoctoral researchers, and graduate students) and disciplines (psychology, political science, and sociology; see *Method*) about their likelihood of adopting validated AI-based tools for predicting experimental effects across various applications. The three scientific applications evaluated above ranked among the top four for which respondents expressed the greatest likelihood of using LLMs. Specifically, researchers reported (see Figure 3D) being most likely to use AI-based tools for power analysis ($M = 65.9$; $SD = 32.1$), intervention selection ($M = 58.4$; $SD = 31.7$), pilot testing ($M = 57.5$; $SD = 32.9$), and identifying published effects in need of replication ($M = 51.7$; $SD = 31.5$). Overall, 76.30% of respondents indicated a greater than 50% likelihood of using AI for at least one of the listed applications (See SI 3.5 including Figures S6-S9 for more results). Taken together, our analysis of potential scientific uses and social scientists' reported likelihood of adoption suggests that LLMs hold potential to support scientific advancement.

Assessing risks in LLM predictions

While LLM predictions of experimental results have several scientific uses, they also pose risks that warrant careful consideration. Below, we outline key concerns, present survey data from 460 social scientists on the extent to which social scientists are concerned about each (101-point scale; 0 = Not at all concerned, 100 = Extremely concerned; See Figure S8 in SI), and empirically assess two risks using our test archives.

A prominent concern is that LLM predictions may exhibit systematic biases against groups that are misrepresented or underrepresented in training data, yielding stereotype-consistent or inaccurate responses [8, 28, 54, 55]. In our survey, concerns regarding stereotyping ($M = 80.0$, $SD = 22.9$) and biases in accuracy ($M = 77.0$, $SD = 24.5$) ranked the highest. Below, we assess whether predictive accuracy varies across demographic groups in the U.S. Survey respondents also reported high concern regarding LLMs' tendency to underestimate the variability of human

responses ($M = 76.8$, $SD = 23.6$) [5, 28, 55, 56], posing a challenge for computing standardized effect size metrics (e.g., Cohen’s d) essential for power analysis. Further, scholars warn that the inner workings of LLMs remain largely opaque, particularly for proprietary models that do not disclose their training data or development processes [5, 27, 57], which could make it difficult to account for unintended influences on LLM predictions, such as model updates or the over-representation of certain topics in the training data. Respondents noted concerns about reproducibility ($M = 74.9$, $SD = 25.3$) and interpretability ($M = 72.4$, $SD = 26.5$). Survey respondents also expressed concern about overreliance on LLMs [27], including simulations replacing human samples altogether ($M = 72.6$, $SD = 28.3$) and unintended epistemic risks, such as diverting research toward more predictable domains ($M = 66.9$, $SD = 26.7$). Sharing these concerns, we view LLM predictions as supplementary tools that can assist—but never replace—the study of human responses. Finally, respondents noted risks of misuse for harmful messaging ($M = 64.3$, $SD = 29.9$). Below, we empirically evaluate two of the risks described above.

Biases in LLM predictions

Given growing concerns about biases in LLM output [8, 54, 55], we used our primary test archive to evaluate LLM performance across subgroups in the U.S. Our approach of prompting LLMs with specific demographic profiles allows us to compare (a) LLM-derived predictions for specific subgroups, with (b) observed effects estimated for those subgroups in the original experiments. Prior evidence of lower accuracy for underrepresented groups in LLM survey predictions [8, 28] suggests a similar risk for experimental predictions.

Importantly, a complete assessment of bias requires substantial treatment-effect heterogeneity across demographic groups to test whether LLM predictions accurately capture between-group differences. However, consistent with prior work [19], original effects in our archive were largely homogeneous, with only 7.7%, 8.8%, and 18.3% of effects significantly moderated by gender, ethnicity, and party, respectively. Given a 5% false positive rate, this implies virtually no heterogeneity across groups, and high predictive accuracy in our archive could simply reflect generalized predictions that ignore subgroup-specific differences.

LLM-derived predictions were generally strongly correlated with original effects across demographic subgroups (See Extended Data Figure 3), but LLM predictions consistently overestimated effect size. We did not find a significant difference in the accuracy of predictions made for women ($r = 0.83$ [0.79, 0.87]; $r_{adj} = 0.90$ [0.87, 0.93]; $RMSE_{adj} = 11.61pp$; $b = 0.56$) and men ($r = 0.79$ [0.73, 0.83]; $r_{adj} = 0.90$ [0.87, 0.93]; $RMSE_{adj} = 11.68pp$; $b = 0.53$) ($p_{r_{adj}} = 0.515$; $p_{RMSE_{adj}} = 0.291$).

While accuracy was generally high for Democrats ($r = 0.72$ [0.65, 0.77]; $r_{adj} = 0.85$ [0.80, 0.89]; $RMSE_{adj} = 16.10pp$; $b = 0.47$) and Republicans ($r = 0.74$ [0.67, 0.79]; $r_{adj} = 0.87$ [0.83, 0.91]; $RMSE_{adj} = 12.87pp$; $b = 0.54$), we found a small but significant difference in RMSE favoring Republicans ($p_{RMSE_{adj}} = 0.001$) but no difference in correlation ($p_{r_{adj}} = 0.277$).

Similarly, predictive accuracy was high for Black ($r = 0.62$ [0.53, 0.69]; $r_{adj} = 0.86$ [0.79,

0.91]; $RMSE_{adj} = 14.79pp$; $b = 0.40$) and white ($r = 0.84$ [0.80, 0.88]; $r_{adj} = 0.90$ [0.87, 0.93]; $RMSE_{adj} = 10.63pp$; $b = 0.60$) participants. Note that while the raw correlation is low for Black participants, this is largely driven by the smaller Black samples used to estimate effects in the original studies (average $N_{Black} = 179.5$, relative to $N_{white} = 1169.7$). Smaller samples lead to greater sampling variability in the estimated effects which in turn undermines the raw correlation coefficient. After adjusting for this variability by correcting for measurement error, we observe relatively high correlations across both groups. Nonetheless, we observe a small but significant difference in both correlations and RMSE favoring white participants ($p_{r_{adj}} = 0.044$; $p_{RMSE_{adj}} < .001$).

Overall, while we find relatively high correlations across demographic groups, we find small differences in accuracy favoring white, and Republican, samples. Nevertheless, given the uniformity in the original effects of our sample, experimental archives with greater heterogeneity in actual effects would be critical to provide a more comprehensive investigation of bias.

Risk of harmful use

We next consider whether LLMs can be used to predict the results of experiments on socially harmful outcomes. While this capacity has positive applications (e.g., identifying harmful content for content moderation, designing counter-messaging strategies), it could be misused to increase the impact of harmful content (e.g., anti-vaccination messaging campaigns; [58]). Although publicly-available LLMs incorporate first-order guardrails to prevent *generating* harmful content (e.g., refusing prompts like “write an effective message to reduce trust in COVID-19 vaccines” [59]), these guardrails may not prevent the use of LLMs to *identify* effective harmful content from a set of options.

We assess this risk using data from a recent experiment [43] that measured the impact of vaccination-related Facebook posts on reducing COVID-19 vaccination intentions. GPT-4-derived predictions were meaningfully correlated with effect size estimates ($r = .49$; $r_{adj} = .96$ [.71, 1.0]) [43]. The five posts selected using LLM predictions as having the most negative effects on vaccine intentions were estimated in the original study to reduce vaccine intentions by 2.77pp, indicating that LLMs could be misused for identifying harmful content (See SI, Figure S10). Our findings indicate that currently implemented first-order guardrails may be insufficient for preventing misuse and suggest a need for “second-order” guardrails to limit the use of LLMs to simulate the effects of harmful messaging. To help avoid potential antisocial applications, we shared these findings with relevant departments at leading AI companies prior to publicly posting this paper.

Discussion

Identification of causal relationships via controlled experiments is fundamental to social and behavioral science. Across our large primary test archive - containing 469 experimental effects involving 119,330 human participants - we find that predictions derived from both proprietary

and open-weight LLMs were strongly correlated with original experimental effect sizes. The correlation remained high for experimental effects that could not have been in the training data, suggesting that LLMs provide meaningful value in forecasting previously unknown experimental results. Further, LLM-derived predictions matched the accuracy of expert predictions in our secondary archive of 606 treatment effects from 15 megastudies testing large-scale social science interventions. Notably, despite high correlations, indicating LLM-derived predictions explain meaningful variance in the effect sizes and can distinguish among effects of differing magnitudes, LLM-derived predictions systematically over-estimated effect sizes, as reflected in the coefficients from regressing original on predicted effect sizes. This pattern underscores that scaling LLM-derived predictions requires re-calibration when applying them to predict experiments in specific empirical settings. Interestingly, this pattern parallels overestimation of absolute effect sizes by lay human forecasters in our primary archive, expert forecasters in our secondary archive, and expert forecasters in recent work [42]. Nevertheless, given that our approach did not involve any specialized optimization (e.g., fine-tuning, prompt filtering), future work may be able to improve predictive performance.

While LLM predictions were generally highly correlated with original effect sizes, we observed some variations. Correlations were lower for experiments targeting stable attitudes and those yielding small effects. In our secondary archive, correlations were higher for text-based experiments with survey outcomes and weaker for non-text treatments and field-based behavioral outcomes. Notably, these differences in LLM performance broadly mirrored patterns in human forecasts, suggesting that some experimental effects may be harder to predict. Further, while correlations were generally high across demographic groups, we found small differences favoring white, over Black, samples, and Republican, over Democratic samples. Moreover, by necessity, our analysis required several analytic decisions. We examined some (e.g., ensemble prompting) and found they shape correlations, but other untested choices (e.g., prompting models to simulate specific demographic profiles) may have contributed to predictive accuracy, and deviations from our approach could reduce performance.

Our findings point to several scientific applications of interest to social scientists, as reflected in a large survey. First, LLMs offer a low-cost tool for efficiently pilot testing ideas before pursuing them in experiments, potentially supporting the development of general theories of human behavior from large amounts of causal effects [60, 61]. Second, LLM predictions can help assess the robustness of effects estimated from underpowered samples, helping to prioritize findings that warrant replication with high power samples; future work could evaluate this application using large-scale databases of published effects (e.g., SCORE; [62]). Third, LLM predictions can improve selection of effective interventions targeting socially desirable outcomes, especially in time-sensitive contexts (e.g., public health emergencies) where rapid and low-cost evaluation of many messages could help identify promising candidates for human testing. Beyond these applications, LLM predictions could support power analyses, serve as Bayesian priors, or approximate results where human experiments are ethically constrained (e.g., misinformation research) [2]. Across applications, the largest gains come from combining LLM predictions with existing approaches, underscoring their role as supplementary tools, not

substitutes for human samples [4, 26, 27, 63].

Although our study provides an initial benchmark for the validity of LLM-derived predictions of experimental results, more research is needed to address remaining questions, limitations, and risks. First, our archive is not well suited for detecting differential predictive accuracy across demographic groups because its treatment effects were largely homogeneous. Future research should examine archives with greater heterogeneity, additional demographic dimensions (e.g., education levels), intersectional groups (e.g., Black women), and non-US samples [5, 8]. Second, more research is required to identify factors shaping predictive accuracy, including research domain (e.g., studies related to health), outcome type (e.g., self-reported vs. behavioral outcome), presence of related study materials or scientific literature in training data, or prompt wording. We find that LLMs systematically overestimate treatment effects and show that re-calibrating the scale of predictions improves accuracy. In practice, recalibration should be estimated from related benchmark datasets wherever possible. Without such benchmark data, LLM predictions should be taken as estimates of relative effect size. Our re-calibration method is tailored to our archive and may not generalize to substantially different experimental effects. More broadly, prompting and analytical choices (e.g., simulating individual demographic profiles) may influence accuracy in non-obvious ways (see SI 3.2), warranting systematic evaluation [64]. Third, we have focused on estimating treatment effects on the studies’ original outcome scales, rather than on standardized scales (e.g., Cohen’s d), which are important for applications like power-analysis. Prior work shows that LLMs underestimate the variance of human responses [28, 56], complicating standardized effect size estimation and statistical inference. Fourth, black-box, proprietary models may limit reproducibility due to opaque training data [5, 27, 57]; future studies should assess reproducibility over time and across models, which ideally requires documented information about models’ training data cutoffs and continuous sources of experimental data that are outside of these models’ training data. Without comprehensive validation, deploying these models risks scientific harm, including perpetuating bias [65], prioritizing ideas based on unreliable predictions, and incentivizing more predictable studies [27]. This highlights the need for research on the long-term effects of AI use in science.

Ultimately, the ability to simulate experiments and generate relatively accurate predictions at low cost, could advance scientific and practical goals. Using AI to augment and inform, rather than replace, human judgment [21, 63] can help the social and behavioral sciences benefit from emerging technologies while remaining grounded in the fields’ collective values. While we view this work as an initial investigation, we anticipate that future research, such as efforts to improve accuracy (e.g., model fine-tuning), validate performance across contexts (e.g., non-U.S. datasets, field experiments), and mitigate risks (e.g., reducing bias, modeling variance), will be critical for advancing the use of LLMs to meaningfully support theoretically grounded, reliable, and practically useful social and behavioral science.

Method

Primary archive. Our primary testing archive of survey experimental results is composed of 50 experiments conducted between 2016 and 2022 by TESS, an NSF-sponsored program in which researchers - typically social scientists - propose survey-based experiments that, if selected through a peer-review process, will be run on nationally representative probability samples. We supplemented the archive with 20 experiments reported in a meta-analysis [19] of two recent replication projects [66, 67]. The meta-analysis tested experimental results obtained from probability samples (studies conducted prior to 2012 largely via TESS) in new experiments they conducted on convenience samples (i.e., Amazon MTurk) and found similar results across the two types of samples. We included data collected through both sample sources to maximize statistical power. TESS requires study authors to submit proposals stating their hypotheses and justifying their sample size, which are then peer-reviewed prior to fielding the experiment on nationally representative samples, effectively providing us an archive of pre-registered, highly powered, high-quality experiments with openly posted stimuli and data. We obtained the final archive of 70 experiments after excluding studies that did not meet our inclusion criteria of experiments (a) which target the general U.S. population, (b) which do not use a conjoint or list design, (c) whose stimuli were primarily or entirely text-based, (d) wherein randomization did not depend on participants' responses to questions. We considered studies to be text-based if they primarily relied on text and any information communicated via images was redundant with the text. For each study in the archive, we included up to 12 conditions and three outcome items corresponding to the first three outcome variables measured post manipulation. In addition to coding the original studies' manipulation text and outcome questions, the authors along with a team of 8 research assistants additionally coded metadata, including the study's hypotheses, discipline(s), topic, and publication status, via a process involving multiple rounds of verification to ensure high accuracy. We considered unpublished studies, and studies published or posted as preprints after September 2021, unlikely to have been included in GPT-4's training data based on archived OpenAI documentation <https://archive.is/WMcad#selection-1835.0-1835.5> (archived 23 Nov 2024, following when we recorded LLM responses) and GPT-4's technical report [68]. In the SI, see *Primary experimental archive details* for more information on this archive, including our criteria for including experiments, experimental conditions, and outcome items, and the coding process, and Table S3 in the SI for a full list of its experiments.

This primary testing archive has several strengths. First, the experiments are high quality: all are highly statistically powered, peer-reviewed, and conducted on nationally representative samples, with preregistered study designs and materials, as well as openly accessible materials and data. The use of nationally representative samples of Americans is particularly valuable, allowing us to assess the accuracy of LLM-derived predictions for demographic subgroups. Second, the archive is substantively broad and diverse. The experiments were designed by 77 social and behavioral scientists from a range of fields (e.g., political science, psychology, sociology, social policy, public health, communication) and test the effects of many different types of experimental treatments (e.g., framing effects, salience of topics, priming social identities) on

a range of outcomes (e.g., political, cultural, and religious attitudes, prejudice towards minorities, happiness) (see Table S3 in the SI for the full list of experiments). Third, results for a large number of experiments were not published, nor posted publicly, by the end of GPT-4’s training data window, allowing us to specifically test for LLMs’ predictive capacity on experiments that the model could not have been exposed to. Fourth, rather than rely on the original researchers’ analyses, we implement a consistent analytic approach to estimate experimental treatment effects. Doing so allowed us to avoid researcher bias and also to estimate all possible experimental contrasts, including those effects not hypothesized by the original researchers and therefore unlikely to have been reported in published or publicly posted papers. Finally, we collected forecasts from 2659 workers on Prolific (49.4% women, 48.9% men, 1.6% other; $Mage = 41.73$; 70.4% White) for all the experiments in our primary test archive (see *Layperson Forecasts* and Figure S1, SI), which provided us a benchmark against which we compared LLM performance.

The primary testing archive also has important limitations. Most critically, it only features studies representative of the US population. Also, while it includes studies from a diversity of disciplines, many disciplines are not included (e.g., cognitive psychology, behavioral economics, development economics, marketing). Finally, the archive consists entirely of survey experiments with text-based stimuli and self-reported dependent measures, and does not include field experiments, behavioral dependent variables, or image or video stimuli.

Primary archive: Analytical approach. In the primary archive, we used a consistent analytical approach across experiments. To assess predictive accuracy, we (a) randomly selected one condition per study to serve as the reference condition against which all others are compared (and one dependent variable, in cases where the original study contained multiple), (b) computed the LLM-predicted treatment effects using the approach described above, (c) correlated the resulting predicted treatment effects with actual effects estimated using the original data, and (d) repeated this process 32 times (to account for randomness in step (a)), recording the median correlation coefficient. Note that we were unable to pre-register our analytic approach because of our familiarity with the analyzed test archives, but we present several robustness check analyses (see SI 2.1, 2.2, and 3.1).

In addition to raw correlations (r), we report adjusted correlations (r_{adj}) that adjust for statistical uncertainty in estimates of original treatment effects. To compute these error-adjusted correlations, we used a two-stage random effects meta-analysis to account for the statistical uncertainty in the treatment effects from each original study. For the first stage, we estimated treatment effects using OLS regression, extracting the corresponding covariance matrix of the sampling errors. For the second stage we assumed that predicted- and true- effects follow a bivariate normal distribution, and estimate the parameters of this distribution incorporating the aforementioned sampling errors using the `metafor` R package [69]. The estimated correlation parameter ρ of this distribution, and its 95% confidence interval, is the disattenuated or adjusted correlation (r_{adj}). We also calculated the absolute accuracy of predictions by estimating the RMSE, adjusting for the uncertainty in treatment effects (θ) using the following formula, similar

to [41]:

$$\text{Adjusted RMSE} = \sqrt{\frac{1}{N} \sum_i (y_i - x_i)^2 - \theta_i^2} \quad (1)$$

Secondary archive of intervention megastudies. We additionally analyzed 15 recent, large-scale, many-treatment experiments testing interventions crafted by researchers from an array of disciplines including psychology, political science, marketing, and economics, sociology, organizational behavior, as well as non-academic practitioners. We included megastudies targeting socially beneficial outcomes that were conducted on sufficiently large samples: (a) Allen et al. (2024; N = 9,228) testing the effects of 90 messages on vaccination intentions, (b) Voelkel et al., (2025; N = 10,638) testing the effects of 10 widely cited treatments on climate change attitudes, (c) Dellavigna & Pope (2018, N = 9,321) testing 15 treatments targeting effortful task behavior, (d) Vlasceanu et al. (2024, N = 6,735) testing 11 text treatments on climate change attitudes, (e) Tappin et al. (2023, N = 62,738) testing 59 treatments targeting UBI and immigration policy attitudes, (f) Zickfeld et al. (2025, N = 20,553) testing 21 treatments targeting tax compliance, (g) Mason et al. (2025, N = 11,733) testing 10 treatments targeting voting intent, (h) Goldwert et al. (2026, N = 27,343) testing 17 treatments targeting climate action, (i) Voelkel et al. (2024, N = 25,121) testing 25 treatments targeting political beliefs, (j) Broockman et al. (2024, N = 61,869) testing 172 treatments targeting various political attitudes, (k) Saccardo et al. (2025, N = 259,249) testing 14 treatments targeting COVID vaccine booster uptake, (l) Milkman et al. (2022, N = 662,170) testing 22 text message treatments on behavioral vaccine uptake. (m) Dellavigna & Linos (2022, N = 960,472) a meta-analysis of 14 nudge treatments targeting several behavioral outcomes. (n) Milkman et al. (2024, N = 2,677,402) testing 7 treatments targeting COVID vaccine booster uptake, (o) Milkman et al. (2021, N = 57,790) testing 53 28-day-long treatments targeting behavioral gym attendance. See *Secondary archive of megastudies*, *Expert forecasts*, and Table S4 in the SI for more information.

Secondary archive: Analytical approach. Given that the megastudies always included a clearly defined control group against which the impact of all treatments were computed, unlike in the primary archive, our goal here was to assess how accurately our LLM-based approach can distinguish the most- and least- effective interventions in a megastudy (e.g., [41]). For this reason, we estimated the correlation for all treatment effects (relative to the control condition) *within a single study*, and then calculated a meta-analytic mean of these correlations. We computed adjusted correlations (r_{adj}) using the approach described above for the primary archive. Note that RMSE is not reported for this archive because the wide variation in outcome variable scales across experiments (e.g., Likert scale measure of climate change attitudes versus number of days someone went to the gym) makes it an unreliable metric for cross-experimental comparison.

LLM Prompting. Our approach involved directly prompting the LLM API, rather than using commercial web chat applications (e.g., ChatGPT). ChatGPT - a chatbot designed for broad commercial use - has additional capabilities, such as internet access, that the GPT-4 model itself does not have. LLM prompts were administered in unique sessions with no memory across sessions, ensuring each response was generated independent of the others. Automating the

process of obtaining and recording LLM responses also helped us avoid manual coding errors. Our primary analysis uses GPT-4, which we estimated to be the most advanced model when we first submitted this paper. Due to practical constraints such as rate limits and costs, we restricted our analysis to this one proprietary model. In addition to GPT-4, for our main analyses, we also use prominent open-weight models Gemma-3, GPT-OSS, and Deepseek-v3, and believe that these models' responses will remain reproducible even if proprietary models are changed in non-transparent ways.

Broadly speaking, to achieve the goal of predicting experimental results, LLMs could be prompted either to (a) directly predict experimental results [40], similar to what forecasters are required to do, or (b) simulate individual participants' responses to experimental stimuli, similar to what research participants do. Here, we adopt the latter strategy because simulation more closely mirrors the original data-generating process and avoids the meta-level judgments required by direct forecasting. This approach also allowed us to examine accuracy for demographic subgroups. Note that for each prompt, we sampled five LLM responses and averaged them.

In all prompts, we ordered the scale labels of outcome scales such that higher values on the scale were paired with labels indicating positive valence or higher degree of the measured quantity (e.g., 1 = Not at all to 5 = Very much, rather than 1 = Very Much to 5 = Not at all). While most original treatments in the secondary archive relied on text, some treatments (e.g., [47]) had to be converted to static text (e.g. transcripts of videos and high-level descriptions of image content) before being added to prompts. For megastudies with behavioral outcomes (e.g., vaccine uptake), the prompt asked how likely the hypothetical participant would be to engage in the particular behavior. See *Prompting strategy* in SI for further detail.

Survey of social scientists. To obtain a large and diverse sample of social scientists, in May 2025, we distributed the survey via email to researchers who (a) have submitted experimental ideas to TESS, (b) were contacted for previous forecasting or metascience projects [47, 70], and (c) are members of the Society for Judgment and Decision Making listserv. Our sample includes 460 participants (40.84% women, 20.95% non-White) representing several social science disciplines (33.5% Psychologists, 31.3% Political Scientists, 17.6% Sociologists), and career stages (20.2% Professors, 20.2% Associate Professors, 21.3% Assistant Professors, 16.3% graduate students, 11.5% postdocs). Additional details about the sample, recruitment strategy, and a list of all questions can be found in the SI.

References

1. Bail, C. A. Can generative AI improve social science? *Proceedings of the National Academy of Sciences* **121**, e2314021121 (2024).
2. Grossmann, I. *et al.* AI and the transformation of social science research. *Science* **380**, 1108–1109 (June 2023).
3. Luo, X. *et al.* Large language models surpass human experts in predicting neuroscience results. *Nature human behaviour* **9**, 305–315 (2025).

4. Crockett, M. & Messeri, L. Should large language models replace human participants? en-us. <https://osf.io/4zdx9> (Jan. 2024).
5. Abdurahman, S. *et al.* Perils and opportunities in using large language models in psychological research. *PNAS nexus* **3**, pgae245 (2024).
6. Argyle, L. P. *et al.* Out of one, many: Using language models to simulate human samples. *Political Analysis* **31**, 337–351 (2023).
7. Bisbee, J., Clinton, J. D., Dorff, C., Kenkel, B. & Larson, J. M. Synthetic replacements for human survey data? The perils of large language models. *Political Analysis* **32**, 401–416 (2024).
8. Atari, M., Xue, M. J., Park, P. S., Blasi, D. & Henrich, J. Which Humans? en-us. <https://psyarxiv.com/5b26t/> (Sept. 2023).
9. Dillion, D., Tandon, N., Gu, Y. & Gray, K. Can AI language models replace human participants? English. *Trends in Cognitive Sciences* **27**, 597–600. ISSN: 1364-6613, 1879-307X (July 2023).
10. Horton, J. J. Large Language Models as Simulated Economic Agents: What Can We Learn from Homo Silicus? en (2023).
11. Binz, M. & Schulz, E. Using cognitive psychology to understand GPT-3. *Proceedings of the National Academy of Sciences* **120**, e2218523120 (Feb. 2023).
12. Bai, H., Voelkel, J. G., Muldowney, S., Eichstaedt, J. C. & Willer, R. LLM-generated messages can persuade humans on policy issues. *Nature Communications* **16**, 6037 (2025).
0. Hackenburg, K. *et al.* The levers of political persuasion with conversational artificial intelligence. *Science* **390**, eaea3884 (2025).
13. (FAIR), M. F. A. R. D. T. *et al.* Human-level play in the game of Diplomacy by combining language models with strategic reasoning. *Science* **378**, 1067–1074 (Dec. 2022).
14. Park, J. S. *et al.* *Generative agents: Interactive simulacra of human behavior* in *Proceedings of the 36th annual acm symposium on user interface software and technology* (2023), 1–22.
15. Schramowski, P., Turan, C., Andersen, N., Rothkopf, C. A. & Kersting, K. Large pre-trained language models contain human-like biases of what is right and wrong to do. *Nature Machine Intelligence* **4**, 258–268 (2022).
16. Yin, Y., Jia, N. & Waksalak, C. J. AI can help people feel heard, but an AI label diminishes this impact. *Proceedings of the National Academy of Sciences* **121**, e2319112121 (Apr. 2024).
17. Lehr, S. A., Caliskan, A., Liyanage, S. & Banaji, M. R. ChatGPT as research scientist: Probing GPT’s capabilities as a research librarian, research ethicist, data generator, and data predictor. *Proceedings of the National Academy of Sciences* **121**, e2404328121 (2024).

18. *Time-sharing experiments for the social sciences*, www.tessexperiments.org <https://www.tessexperiments.org/>.
19. Coppock, A., Leeper, T. J. & Mullinix, K. J. Generalizability of heterogeneous treatment effect estimates across samples. en. *Proceedings of the National Academy of Sciences* **115**, 12441–12446. ISSN: 0027-8424, 1091-6490 (Dec. 2018).
20. Chang, S., Kennedy, A., Leonard, A. & List, J. A. *12 Best Practices for Leveraging Generative AI in Experimental Research* tech. rep. (National Bureau of Economic Research, 2024).
21. Broska, D., Howes, M. & van Loon, A. The mixed subjects design: Treating large language models as potentially informative observations. *Sociological Methods & Research* **54**, 1074–1109 (2025).
22. Chu, J. Y. *et al.* Academics are more specific, and practitioners more sensitive, in forecasting interventions to strengthen democratic attitudes. en. *Proceedings of the National Academy of Sciences* **121**, e2307008121. ISSN: 0027-8424, 1091-6490 (Jan. 2024).
23. DellaVigna, S. & Linos, E. RCTs to Scale: Comprehensive Evidence From Two Nudge Units. en. *Econometrica* **90**, 81–116. ISSN: 0012-9682 (2022).
24. Dreber, A. *et al.* Using prediction markets to estimate the reproducibility of scientific research. en. *Proceedings of the National Academy of Sciences* **112**, 15343–15347. ISSN: 0027-8424, 1091-6490 (Dec. 2015).
25. Milkman, K. L. *et al.* Megastudies improve the impact of applied behavioural science. en. *Nature* **600**, 478–483. ISSN: 1476-4687 (Dec. 2021).
26. Harding, J., D’Alessandro, W., Laskowski, N. G. & Long, R. AI language models cannot replace human research participants. en. *AI SOCIETY*. ISSN: 1435-5655. <https://doi.org/10.1007/s00146-023-01725-x> (July 2023).
27. Messeri, L. & Crockett, M. J. Artificial intelligence and illusions of understanding in scientific research. en. *Nature* **627**, 49–58. ISSN: 1476-4687 (Mar. 2024).
28. Bisbee, J., Clinton, J. D., Dorff, C., Kenkel, B. & Larson, J. M. Synthetic Replacements for Human Survey Data? The Perils of Large Language Models. en. *Political Analysis*, 1–16. ISSN: 1047-1987, 1476-4989 (May 2024).
29. Binz, M. *et al.* A foundation model to predict and capture human cognition. *Nature*, 1–8 (2025).
30. Petrov, N. B., Serapio-García, G. & Rentfrow, J. Limited Ability of LLMs to Simulate Human Psychological Behaviours: a Psychometric Analysis. en. arXiv:2405.07248 [cs]. <http://arxiv.org/abs/2405.07248> (May 2024).
31. Tjuatja, L., Chen, V., Wu, T., Talwalkwar, A. & Neubig, G. Do llms exhibit human-like response biases? a case study in survey design. *Transactions of the Association for Computational Linguistics* **12**, 1011–1026 (2024).

32. Chen, Y., Hu, Y. & Lu, Y. Predicting Field Experiments with Large Language Models. *arXiv preprint arXiv:2504.01167* (2025).
33. Wang, X. *et al.* Self-Consistency Improves Chain of Thought Reasoning in Language Models in *The Eleventh International Conference on Learning Representations* ().
34. Arora, S. *et al.* Ask Me Anything: A simple strategy for prompting language models in *The Eleventh International Conference on Learning Representations* ().
35. Voronov, A., Wolf, L. & Ryabinin, M. Mind your format: Towards consistent evaluation of in-context learning improvements in *Findings of the Association for Computational Linguistics: ACL 2024* (2024), 6287–6310.
36. Hou, B., O’connor, J., Andreas, J., Chang, S. & Zhang, Y. Promptboosting: Black-box text classification with ten forward passes in *International Conference on Machine Learning* (2023), 13309–13324.
37. Collaboration, O. S. Estimating the reproducibility of psychological science. *Science* **349**, aac4716 (2015).
38. Milkman, K. L. *et al.* Megastudy shows that reminders boost vaccination but adding free rides does not. *Nature* **631**, 179–188 (2024).
39. Milkman, K. L. *et al.* A 680,000-person megastudy of nudges to encourage vaccination in pharmacies. en. *Proceedings of the National Academy of Sciences* **119**, e2115126119. ISSN: 0027-8424, 1091-6490 (Feb. 2022).
40. Zickfeld, J. H. *et al.* Effectiveness of ex ante honesty oaths in reducing dishonesty depends on content. *Nature Human Behaviour* **9**, 169–187 (2025).
41. Broockman, D. E., Kalla, J. L., Caballero, C. & Easton, M. Political practitioners poorly predict which messages persuade the public. *Proceedings of the National Academy of Sciences* **121**, e2400076121 (2024).
42. Dellavigna, S. & Vivalt, E. Forecasting Social Science: Evidence from 100 Projects.
43. Allen, J., Watts, D. J. & Rand, D. G. Quantifying the impact of misinformation and vaccine-skeptical content on Facebook. *Science* **384**, eadk3451 (2024).
44. DellaVigna, S. & Pope, D. What motivates effort? Evidence and expert forecasts. *The Review of Economic Studies* **85**, 1029–1069 (2018).
45. Vlasceanu, M. *et al.* Addressing climate change with behavioral science: A global intervention tournament in 63 countries. *Science advances* **10**, eadj5778 (2024).
46. Tappin, B. M., Wittenberg, C., Hewitt, L. B., Berinsky, A. J. & Rand, D. G. Quantifying the potential persuasive returns to political microtargeting. *Proceedings of the National Academy of Sciences* **120**, e2216261120 (2023).
47. Voelkel, J. G. *et al.* Megastudy testing 25 treatments to reduce antidemocratic attitudes and partisan animosity. *Science* **386**, eadh4764 (2024).

48. Voelkel, J. G. *et al.* A registered report megastudy on the persuasiveness of the most-cited climate messages. *Nature Climate Change*, 1–12 (2026).
49. Saccardo, S. *et al.* Field testing the transferability of behavioural science knowledge on promoting vaccinations. *Nature Human Behaviour* **8**, 878–890 (2024).
50. Mason, K. *et al.* A Megastudy of Behavioral Interventions to Increase Voter Registration Ahead of the 2024 US Presidential Election (2025).
51. Goldwert, D. *et al.* A megastudy of behavioral interventions to catalyze public, political, and financial climate advocacy. *PNAS nexus* **5**, pgaf400 (2026).
52. Druckman, J. N. *Experimental thinking* (Cambridge University Press, 2022).
53. Freese, J. & Peterson, D. Replication in social science. *Annual Review of Sociology* **43**, 147–165 (2017).
54. Bai, X., Wang, A., Sucholutsky, I. & Griffiths, T. L. Explicitly unbiased large language models still form biased associations. *Proceedings of the National Academy of Sciences* **122**, e2416228122 (2025).
55. Wang, A., Morgenstern, J. & Dickerson, J. P. Large language models that replace human participants can harmfully misportray and flatten identity groups. *Nature Machine Intelligence*, 1–12 (2025).
56. Park, P. S., Schoenegger, P. & Zhu, C. Diminished diversity-of-thought in a standard large language model. en. *Behavior Research Methods*. ISSN: 1554-3528. <https://doi.org/10.3758/s13428-023-02307-x> (Jan. 2024).
57. Frank, M. C. Openly accessible LLMs can help us to understand human cognition. *Nature Human Behaviour* **7**, 1825–1827 (2023).
58. Wack, M., Ehrett, C., Linvill, D. & Warren, P. Generative propaganda: Evidence of AI’s impact from a state-backed disinformation campaign. *PNAS nexus* **4**, pgaf083 (2025).
59. Response retrieved from ChatGPT-4 on 22 May, 2024, <https://chatgpt.com/share/2a2b66d0-e9fd-421f-bedd-bb6f9354742a> <https://chatgpt.com/share/2a2b66d0-e9fd-421f-bedd-bb6f9354742a>.
60. Almaatouq, A. *et al.* Beyond playing 20 questions with nature: Integrative experiment design in the social and behavioral sciences. *Behavioral and Brain Sciences* **47**, e33 (2024).
61. Zhu, J.-Q., Xie, H., Arumugam, D., Wilson, R. C. & Griffiths, T. L. Using Reinforcement Learning to Train Large Language Models to Explain Human Decisions. *arXiv preprint arXiv:2505.11614* (2025).
62. Alipourfard, N. *et al.* Systematizing confidence in open research and evidence (SCORE). *SocArXiv* (2021).
63. Cremer, D. D. & Kasparov, G. AI Should Augment Human Intelligence, Not Replace It. *Harvard Business Review*. ISSN: 0017-8012. <https://hbr.org/2021/03/ai-should-augment-human-intelligence-not-replace-it> (Mar. 2021).

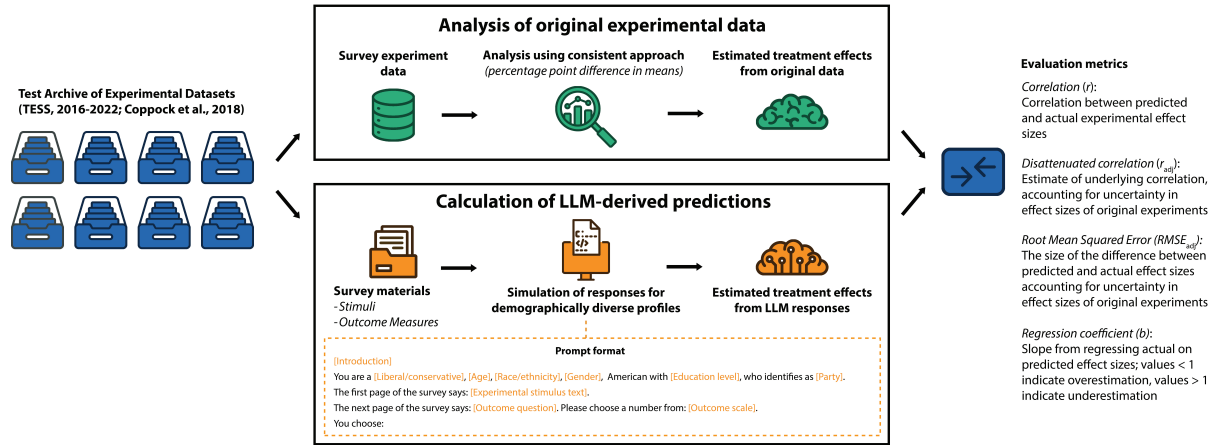
64. Cummins, J. The threat of analytic flexibility in using large language models to simulate human data: A call to attention. *arXiv preprint arXiv:2509.13397* (2025).
65. Obermeyer, Z., Powers, B., Vogeli, C. & Mullainathan, S. Dissecting racial bias in an algorithm used to manage the health of populations. *Science* **366**, 447–453 (Oct. 2019).
66. Coppock, A. Generalizing from Survey Experiments Conducted on Mechanical Turk: A Replication Approach. en. *Political Science Research and Methods* **7**, 613–628. ISSN: 2049-8470, 2049-8489 (July 2019).
67. Mullinix, K. J., Leeper, T. J., Druckman, J. N. & Freese, J. The Generalizability of Survey Experiments. en. *Journal of Experimental Political Science* **2**, 109–138. ISSN: 2052-2630, 2052-2649 (2015).
68. Achiam, J. *et al.* Gpt-4 technical report. *arXiv preprint arXiv:2303.08774* (2023).
69. Viechtbauer, W. & Viechtbauer, M. W. Package ‘metafor’. *The comprehensive R Archive network. Package ‘metafor* (2015).
70. Rios, K., Roth, Z. C. & Langston, J. Academics’ stereotypes about identity-related research: The role of intellectual humility (2023).

Acknowledgements: We thank Niles Egan, Kylie Fuller, Vicky Isarraras, Sebastian Kane, Ben Kinder, Akosua Kissi, Kyleigh McPeck, Shane Muldowney, Chrystal Redekopp, and Tommy Ren for research assistance. We are grateful to researchers who shared data, including James Chu, Stefano Dellavigna, Jamie Druckman, Angela Duckworth, Danielle Goldwert, Neil Malhotra, Katie Mason, Katy Milkman, Kayley Okst, Silvia Saccardo, Madalina Vlasceanu, Jan Voelkel. We also thank Jenny Allen, Mohammad Atari, Chris Bail, Michael Bernstein, Morteza Dehghani, David Broockman, Katie Clayton, Molly Crockett, Deep Ganguli, Danica Dillon, Johannes Eichstaedt, Michael Frank, Noah Goodman, Donald Green, Josh Kalla, Percy Liang, Jay McClelland, David Rand, Reed Orchinik, Nicholas Otis, Steve Rathje, Ben Tappin, Austin Van Loon, Adam Waytz and Dave Willner for invaluable feedback on the project.

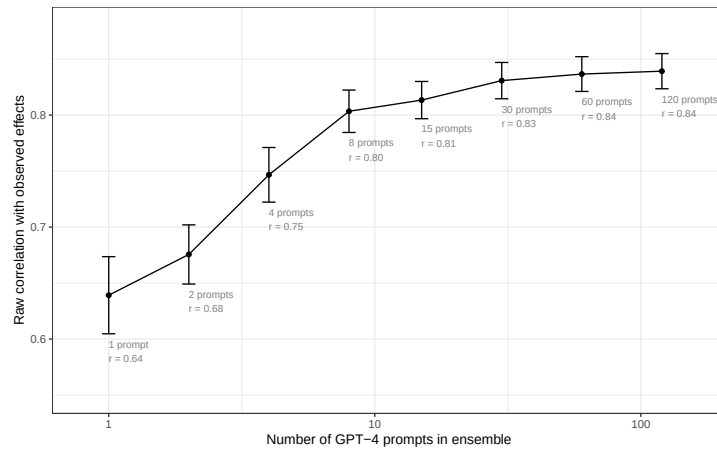
Funding statement: We acknowledge funding from the Stanford Center on Philanthropy and Civil Society.

Author contributions: L.H., A.A., and R.W. designed the research; A.A., L.H., I.G., and R.W. assembled the experimental archives. L.H. and A.A. analyzed the data. A.A., L.H., and R.W. wrote the paper. L.H. built the web tool.

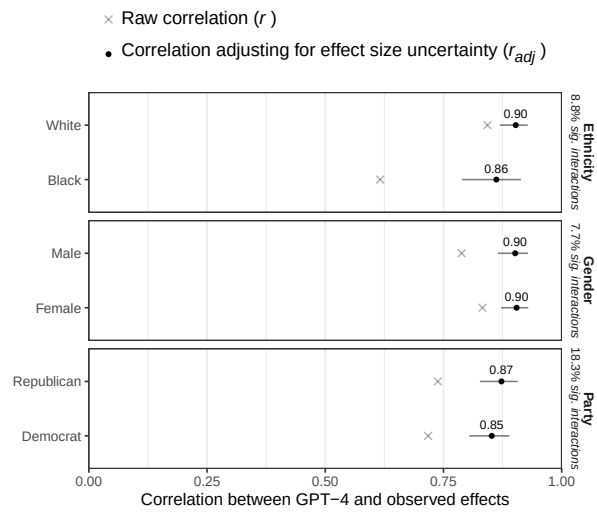
Extended data figures and tables



Extended Data Figure 1: Method. We re-analyze raw data from 70 nationally-representative U.S. studies and estimate treatment effects in a consistent manner. We use a Large Language Model to predict results of those same experiments, providing the original study materials and demographically-diverse participant profiles, and then calculate the average predicted response for each condition. We evaluate the accuracy of the model in terms of the correspondence between observed- and predicted- treatment effects.



Extended Data Figure 2: Averaging across more LLM prompts improves predictive accuracy. Error bars indicate 95% confidence intervals for the mean Pearson correlation across 32 simulation runs. Each run uses a different random subset of prompts (drawn without replacement from 120). The CI reflects uncertainty in the mean across simulation runs, but does not include uncertainty over the sample of 70 experiments or the sample of 120 prompts



Extended Data Figure 3: The accuracy of LLM-derived predictions for demographic subgroups in the US. Error bars indicate 95% confidence intervals.